

Cloud IQ – AI Notes

Introduction

Cloud IQ uses Artificial Intelligence to make cloud cost optimization easier and more effective. During development, we found that simply displaying resource utilization and billing data was not enough. Engineers still had to manually analyze the information and decide what actions should be taken.

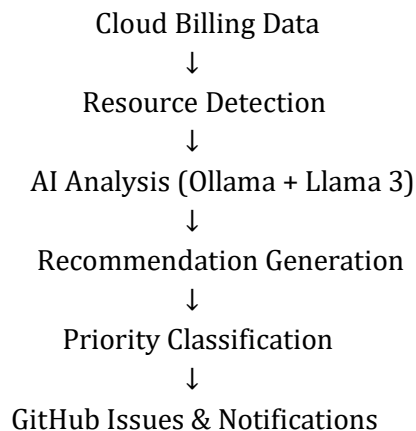
To simplify this process, we integrated Ollama with Llama 3 into the solution. The AI model analyzes cloud resource data, identifies optimization opportunities, and generates recommendations that help users reduce unnecessary cloud spending.

Why AI Was Used

The main purpose of adding AI was to reduce manual effort and improve decision-making. The AI component helps by:

- Identifying underutilized resources
- Detecting potential cloud waste
- Generating optimization recommendations
- Prioritizing findings based on impact
- Providing easy-to-understand explanations

AI Workflow



After a resource is flagged, the AI analyzes its usage patterns and generates recommendations along with a severity level.

AI Features

Resource Analysis

The AI evaluates:

- Resource Type

- CPU Utilization
- Monthly Cost
- Resource Activity

Based on these factors, resources are classified as:

- Idle
- Underutilized
- Oversized
- Efficiently Utilized

Recommendation Generation

Example Input:

- Resource ID: srv-03
- CPU Usage: 2%
- Monthly Cost: \$220

Output:

- Recommended Action: Downsize or Terminate Resource
- Estimated Savings: \$150/month

Priority Classification

P0 – Critical issue requiring immediate attention

P1 – Moderate optimization opportunity

P2 – Minor recommendation

Why Ollama?

- Runs completely offline
- Eliminates API costs
- Provides faster local processing
- Improves privacy and security
- Supports reliable demonstrations without internet access

Reliability

Cloud IQ includes a fallback mechanism to ensure continuous operation. If Ollama becomes unavailable:

- Resource analysis continues
- Rule-based recommendations are generated
- GitHub issues are still created
- Notifications continue to work
- Dashboard generation remains unaffected